Metabolic Power: A Step in the Right Direction for Team Sports

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INVITED COMMENTARY

Metabolic Power: A Step in the Right Direction for Team Sports

Ted Polglaze and Matthias W. Hoppe

The ability of the "metabolic power" model to assess the demands of team-sport activity has been the subject of some interest—and much controversy—in team-sport research. Because the cost of acceleration depends on the initial speed and the costs of acceleration and deceleration are not "equal and opposite," changes in speed must be accounted for when evaluating variable-speed locomotion. The purpose of this commentary is to address some of the misconceptions regarding "metabolic power," acknowledge its limitations, and highlight some of the benefits that energetic analysis offers over alternative approaches to quantifying the demands of team sports.

Keywords: time–motion analysis, player tracking, energetics, acceleration, deceleration

Metabolic power (P_{met}) has been proposed as a tool to estimate the energetic demands of variable-speed locomotion typically seen in team sports.[1] From the outset, it should be stated that this model is not able to fully account for the physical demands of team-sport activity,[2,3] but nor does it purport to. Given existing technological and analytical limitations, together with the complex nature of team sports, the direct and complete measurement of energy demands appears to be a distant dream. Until that becomes possible, the use of a model may help improve our understanding of team-sport energetics. In this brief commentary, we will address some of the misconceptions surrounding P_{met} and highlight both its merits and limitations regarding team-sport applications.

Background

The conceptual basis of the P_{met} model emerges from the biomechanical equivalence between accelerated running on level ground and constant-speed running on an incline.[4] Although this is portrayed as the mutual forward lean of the runner in each situation, it is more correctly described as the similar vectorial distribution of propulsive and gravitational forces manifest in each mode of locomotion.[4] The energy demands of brief, nonsteady-state—and often high-intensity—accelerations cannot be directly measured. However, this is possible for steady-state incline running, where energy cost increases with slope but, as per running on level ground, is independent of speed at a given slope.[5] Accordingly, instantaneous P_{met} can be estimated from the product of speed and energy cost, even for very-short and high-intensity accelerations, thus giving rise to the P_{met} model. Of note, this "equivalent slope" approach has recently been validated with experimental evidence across an extended range of inclines, and shown to be applicable for both acceleration and deceleration.[6]

An alternative P_{met} model exists, derived from the mechanical determinants of energy expenditure (EE) during running[7]; however, this commentary will focus on the original and more prominent version.[4] Both models derive P_{met} solely from speed data, using the absolute value at a given time point together with the magnitude and direction of change from the preceding time point. Hence, P_{met} can be readily obtained from contemporary player-tracking technology, provided sufficient accuracy[8] and sampling rate[9] for instantaneous speed are evident. The series of calculations to derive P_{met} from speed is provided in Table 1.

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Definition, Nomenclature, and Units of Measurement

Common questions regarding P_{met} include "Why is it called 'metabolic' power?" and "Why is it expressed in watts and relative to body mass?" P_{met} has been defined as "the required rate of ATP hydrolysis . . . to perform the muscular work done during running" (p. 1361).[7] Accordingly, it represents the net metabolic demands incurred by locomotion, rather than a measure of mechanical power output. Calculated as the product of instantaneous energy cost (expressed in $J{\cdot}kg^{-1}{\cdot}m^{-1}$) and speed ($m{\cdot}s^{-1}$), P_{met} is thus the amount of energy required per second per kilogram, which can be further reduced to $W{\cdot}kg^{-1}$. The expression of P_{met} in SI—rather than arbitrary—units demonstrates its connection to fundamental principles of measurement. Furthermore, P_{met} can also be expressed as a rate of oxygen demand. Assuming a respiratory quotient of 0.96 for intermittent activity and corresponding energy equivalent of 20.9 kJ per liter of O_2, 1 $W{\cdot}kg^{-1}$ converts to 2.87 $mL{\cdot}kg^{-1}{\cdot}min^{-1}$.[10] Theoretically, mechanical power could be derived from instantaneous speed data and body mass via the equations for force (mass × acceleration) and power (force × velocity), but this simplistic mechanical approach overlooks the physiological processes involved in changing speed, particularly the differences between concentric and eccentric muscular work.[11]

Staying Within the Boundaries: What Metabolic Power Can and Cannot Assess

Although P_{met} is useful for the assessment of the physiological demands of team-sport locomotion, it is not an estimate of overall EE. Nor is it an estimate of mechanical work. It does not purport to quantify the neuromuscular strain imposed by activities such as high-speed running, rapid decelerations, or collisions. These demanding

Table 1 Sample Spreadsheet Showing the Series of Calculations to Derive Metabolic Power From Instantaneous Speed and Actual Formulas for an Excel Spreadsheet

Time, s	Speed, m·s^{-1}	Acceleration, m·s^{-2}	α, degrees	g, m·s^{-2}	Equivalent slope	Equivalent mass, kg	Energy cost, J·kg^{-1}·m^{-1}	Metabolic power, W·kg^{-1}
0.0	1.624							
0.1	1.631	0.070	89.591	9.810	0.007	1.000	4.827	7.872
0.2	1.711	0.800	85.338	9.843	0.082	1.003	7.085	12.122
0.3	1.847	1.360	82.107	9.904	0.139	1.010	9.213	17.017
0.4	1.888	0.410	87.607	9.819	0.042	1.001	5.801	10.951
0.5	1.952	0.640	86.267	9.831	0.065	1.002	6.537	12.761
0.6	2.012	0.600	86.500	9.828	0.061	1.002	6.405	12.886
0.7	2.122	1.100	83.602	9.871	0.112	1.006	8.185	17.369
0.8	2.187	0.650	86.209	9.832	0.066	1.002	6.571	14.370
0.9	2.167	−0.200	−88.832	9.812	−0.020	1.000	4.157	9.009
1.0	2.070	−0.970	−84.353	9.858	−0.099	1.005	2.803	5.801

Abbreviations and equations:

Acceleration (Acc)	(Speed$_t$-Speed$_{t-1}$)/0.1 (NB—10 Hz data divided by 0.1 to determine acceleration value per second)
α	DEGREES(ATAN(9.81/Acc))
g'	SQRT(Acc^2+9.81^2)
Equivalent slope (ES)	TAN(0.5*PI()-ATAN(9.81/Acc))
Equivalent mass (EM)	g'/9.81
Energy cost (EC)	IF(Acc = 0,3.6*1.29,(155.4*ES^5-30.4*ES^4-43.3*ES^3+46.3*ES^2+19.5*ES+3.6)*EM*1.29) (where 3.6 = EC for constant-speed running on level compact terrain and 1.29 = multiplier for grassy terrain)
Metabolic power	EC*Speed

Note: For a detailed explanation of the equations, see di Prampero et al[4] and Osgnach et al.[1]

elements of team-sport activity must be assessed by alternative measures.[7] As with existing displacement methods, P_{met} does not account for numerous other aspects of team-sport activity, including jumping, tackling, and skill-based actions.[1] Even in locomotor activity, P_{met} is unable to quantify the additional energy cost imposed by changing direction (see below for clarification), lateral/backward movement, and unconventional gaits and postures.[12] Notably, these limitations also apply when speed and distance are used to indicate the intensity and volume, respectively, of team-sport activity, although this is seldom acknowledged.

What P_{met} does do is address one aspect of team-sport locomotion overlooked by conventional speed-based approaches—the (substantial) energetic cost of changing speed. While contemporary time–motion analyses do report acceleration characteristics,[13,14] speed and acceleration are usually considered separately, whereas it is the interaction between these 2 parameters that determines energetic demands.[4] The beauty of P_{met} is that it incorporates the magnitude, duration, and sign (ie, positive or negative) of acceleration, along with the initial speed, to provide estimates of instantaneous and cumulative energy demand for variable-speed locomotion, equating to measures of intensity (P_{met}) and volume (EE), respectively, for team-sport activity.

Getting Up to Speed on Acceleration

It is generally accepted that continually changing speed represents one of the most demanding aspects of team-sport activity. From an energetic perspective, there are 2 important considerations when assessing acceleration and deceleration.

Not All Equal Accelerations Are Equal

Although it is conventional in team-sport research to classify accelerations using fixed absolute values,[14,15] this ignores the fact that high accelerations cannot be attained from an elevated starting speed.[16] This was demonstrated in soccer, where 98% of "maximal" accelerations (>2.78 m·s^{-2}) commenced from a starting speed below the threshold for high-speed running, and 85% of these did not reach this threshold.[13] This should not be interpreted as the athlete's not being able to accelerate "maximally" at high speed; rather, the maximal acceleration that can be reached decreases as initial speed increases.[16] Likewise, the magnitude of acceleration decreases as speed increases within an acceleration effort, despite exertion remaining high throughout. It is apparent therefore that acceleration must be interpreted with respect to the accompanying speed. This is perhaps the most beneficial feature of the P_{met} model, in that it is derived from the instantaneous interaction between speed and acceleration. Data from an international hockey match (Figure 1) reveal how accelerations of similar magnitude yield different P_{met} values depending on starting speed.

Sign Matters

Typically, accelerations and decelerations of the same magnitude are deemed equally demanding in team-sport research.[15] However, while deceleration imposes a substantial neuromuscular load, the metabolic cost of slowing down is negligible, even compared with constant-speed running. Reverting to the incline analogy, running uphill raises heart rate and ventilation, whereas exertion on the way

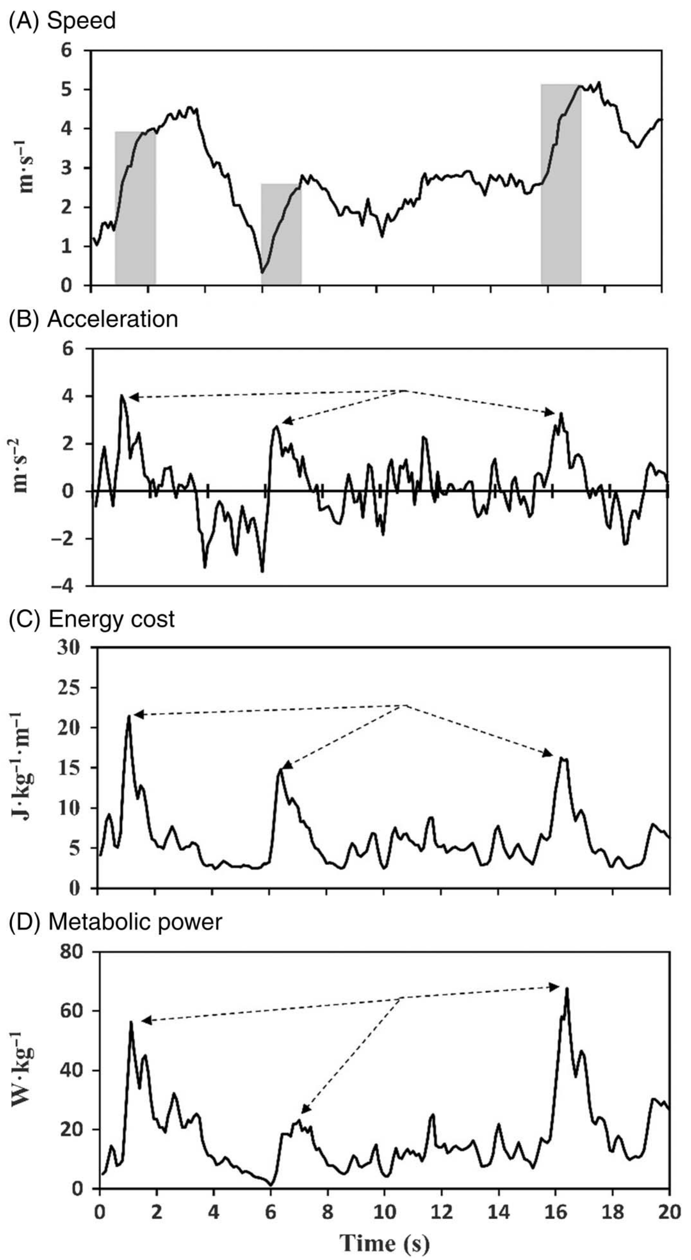

Figure 1 — Twenty seconds of player-tracking data from a male hockey player in an international match. Three discrete accelerations (shaded), all of similar magnitude but with different initial speeds, demonstrate the interaction between speed and acceleration to yield different metabolic-power values. Note also the low energy cost and metabolic power associated with periods of deceleration.

down is minimal but will likely cause muscular soreness the next day. This disparity between the cost of acceleration and deceleration of equal magnitude is partly explained by differences in efficiency between concentric and eccentric muscle contractions[17] due to passive-elastic factors, and highlights the importance of accounting for changes in speed in team sports. If the energy costs were reciprocal, they would effectively cancel each other out over a given period, and average speed would be a suitable indicator of intensity for variable-speed locomotion. However, because the energy cost of deceleration is much less than that of acceleration[4] (see Figure 1), changes in speed must be accounted for when assessing the demands of variable-speed locomotion. This is precisely what P_{met} accomplishes.

Setting Off in a Different Direction

Direction changes occur frequently in team sports,[18] and shuttle running incurs a higher energy cost than straight line running at the same average speed.[19] At face value, the inability to detect and quantify the cost of changing direction would appear to be another limitation of the P_{met} model. However, changing direction comprises a deceleration, an additional force application to redirect the momentum of the body, followed by an acceleration.[20] Changes in speed can be accounted for, so it is only the cost of redirecting the trajectory of the body via postural adjustments, which incidentally is independent of the angle of change,[21] that P_{met} cannot evaluate.

This All Sounds Great in Theory, but Hasn't Metabolic Power Been Shown to Be Invalid?

Several recent studies have suggested that P_{met} is "invalid" due to the weak association between estimated EE and measured oxygen consumption (VO_2) in various aspects of team-sport activity.[2,3,22,23] In using VO_2 as the criterion measure, the underlying premise is that these 2 variables are related, but this is not necessarily the case. While VO_2 measures energy supply, P_{met} is a measure of instantaneous energy demand, and there is no particular reason why these 2 parameters should correspond during nonsteady-state activity, particularly given that VO_2 represents the aerobic contribution—only—to global metabolic demands, including basal metabolism and nonlocomotor activity, whereas P_{met} may be derived from both aerobic and anaerobic sources, but only accounts for the EE attributable to locomotion. Existing validation studies have been unable to account for all sources of energy supply, which—at least to some extent—explains the disparity typically reported between VO_2 and P_{met}. Rather than expecting a close real-time association between these parameters—a condition strictly limited to steady-state aerobic exercise—differences in instantaneous energy supply and demand during variable-speed running may in fact be an important area of future research for team sports. For example, the potential to model aerobic and anaerobic energy contribution[10] represents a novel innovation made possible by this model.

Nevertheless, these previously mentioned studies highlight the limitations of P_{met} in being unable to account for all aspects of team-sport activity. However, they also provide evidence to support the efficacy of P_{met} over traditional methods—particularly when basal metabolic requirements and the lag between energy demand and its supply from aerobic sources are accounted for.[24] In randomized bouts of constant-speed locomotion and a team-sport simulation circuit, P_{met} accurately determined EE for both jogging and running, but accuracy was lower for walking and when changes of direction were included.[2] Even when half of all locomotion was backward, EE was more closely related to P_{met} than to either high-speed running distance or accelerometry measures.[3] Clearly, further research is necessary before the full metabolic cost of team-sport activity can be ascertained, but P_{met} appears to be an improvement on existing approaches. It is noteworthy that the P_{met} model has recently been updated,[25] such that it now differentiates between running and walking (which has a lower and speed-dependent energy cost), and also incorporates the (negligible) cost of air resistance.

Practical Applications: Why Use Metabolic Power in Team Sports?

Common motives for the use of player-tracking systems in team sports include determining the "demands" of competition,[26] detecting the occurrence of fatigue,[27] and assessing the efficacy of various interventions.[28] Invariably, distance and speed are used as indicators of volume and intensity, respectively.[12] However, using a novel approach whereby a conventional speed threshold (14.4 $km{\cdot}h^{-1}$) and comparable P_{met} value (20 $W{\cdot}kg^{-1}$) were adopted, speed-based classification of intensity was shown to underestimate demands in soccer, particularly in training sessions or playing positions associated with less high-speed running,[29] demonstrating that large amounts of high-intensity activity occur at low speed. But, the constant-speed equivalent to 20 $W{\cdot}kg^{-1}$ is in fact 15.5 $km{\cdot}h^{-1}$,[30] meaning that the bias is even more pronounced, as depicted in Figure 2. Regardless, P_{met} represents a more sensitive tool to quantify intensity and therefore can be used to set appropriate individualized thresholds for "critical power"[31] and maximal aerobic power,[10] and to define and detect high-intensity efforts and bouts of repeated high-intensity activity,[32] along with the numerous other purposes for which volume and intensity are evaluated in team sports.

The incongruity between classification of intensity via internal and external load, clearly demonstrated in small-sided games where players obtain high heart rate despite not covering much distance nor reaching high speeds,[33] highlights another aspect of how demands of team-sport activity might be more correctly evaluated using P_{met}. It seems improbable that team-sport athletes only complete <10% of activity at "high intensity" yet spend approximately 60% of playing time above 85% of maximum heart rate.[28] This suggests that the method and/or threshold used to classify external load is dubious, with considerable amounts of high-intensity activity being overlooked. Classification via P_{met} rather than speed yields substantially more high-intensity activity in both soccer[1] and hockey,[30] and has a stronger association with internal load.[31]

Another important application of P_{met} is that it allows high-intensity activity to be expressed in proportion to total EE, rather than the percentage of total playing time and/or distance covered.[7] High-intensity efforts are invariably brief, so neither time nor distance accumulate substantially, whereas adenosine triphosphate (ATP) turnover during this period can be extremely high. The expression of intensity in the energy (rather than time) domain underlines the connection between P_{met} and the supply of ATP for muscular contraction, providing a direct link to the metabolic demands of stochastic team-sport activity.

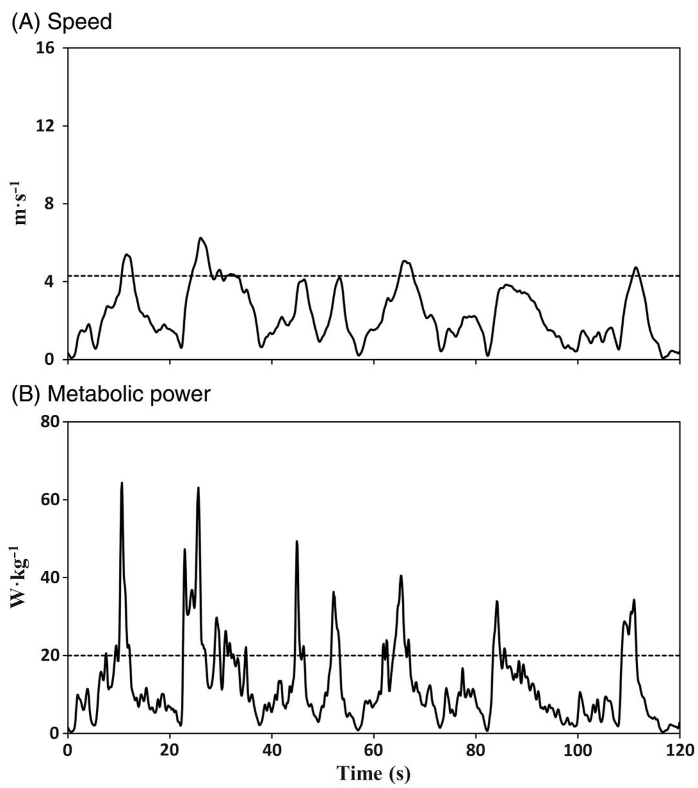

Figure 2 — Differences in classification of intensity between a speed-based threshold (15.5 $km{\cdot}h^{-1} \leftrightarrow 4.3$ $m{\cdot}s^{-1}$) and the equivalent metabolic-power threshold (20.0 $W{\cdot}kg^{-1}$). Data obtained from a male hockey player in an international match. Note that the *y*-axis ranges are adopted in order to place the threshold value in approximately the same relative position.

Conclusion

Metabolic power is not the panacea for team-sport analysis, but it is a very useful tool. By estimating the cost of acceleration in activity comprising perpetual changes in speed, it addresses a fundamental flaw of existing approaches. Consequently, it provides a more comprehensive—yet still incomplete—measure of intensity and volume for variable-speed locomotion, allowing for improved monitoring of training and competition loads and thus represents a step in the right direction in the quest to understand the demands of team sports.

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